

Optimal inventory model with defective units involving controllable backorder rate and variable lead time with mixtures of distribution

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Abstract

This paper considers that the number of defective units in an arrival order to be a binomial random variable. We derive a modified mixture inventory model with backorders and lost sales, in which the order quantity and lead time are decision variables. In our studies, we also assume that the backorder rate is dependent on the length of lead time through the amount of shortages and let the backorder rate is a control variable. In addition, we assume that the lead time demand follows a mixtures of normal distribution, and then relax the assumption about the form of the mixtures of distribution function of the lead time demand and apply the minimax distribution free procedure to solve the problem. Furthermore, we develop an algorithm procedure to obtain the optimal ordering strategy for each case. Finally, two numerical examples are also given to illustrate the results.

Keywords: Defective units; Order quantity; Backorder rate; Minimax distribution free procedure; Mixtures of distribution

1. Introduction

In most of the literatures dealing with inventory problems, either in deterministic or probabilistic model, lead time is viewed as a prescribed constant or a stochastic variable, which, therefore, is not subject to control (see, e.g., Naddor, 1966; Silver and Peterson, 1985). In fact, lead time usually consists of the following components: order preparation, order transit, supplier lead time, delivery time, and setup time (Tersine, 1982). In many practical situations, lead time can be reduced by an added crashing cost; in other words, it is controllable. By shortening the lead time, we can lower the safety stock (SS), reduce the loss

caused by stock-out, improve the service level to the customer, and increase the competitive ability in business.

Liao and Shyu (1991) first presented a probabilistic model in which the order quantity is predetermined and lead time is a unique decision variable. Later, Ben-Daya and Raouf (1994) extended the Liao and Shyu's model by considering both the lead time and the order quantity as decision variables. In their studies (Liao and Shyu, 1991; Ben-Daya and Raouf, 1994), they assume that the probability of allowable stock-out during the lead time is very small, and hence the shortages are neglected. In some recent papers, Ouyang et al. (1996), Ouyang and Chuang (2001) and Wu and Tsai (2001) have also extended Ben-Daya and Raouf's model by considering the stock-out component.

The above body of literature does not describe the possible relationship between the order lot and quantity. As a result of imperfect production of the supplier, and/or damage in transit, it is often that an arrival order lot may contain some defective units. The presence of defective units in orders would have an impact on the number of shortages and the frequency of orders in the inventory system. Therefore, ordering policy determined by conventional inventory models may not be appropriate when applied to the defective inventory situations.

Hence, Ouyang and Wu (1999) consider both lead time and the order quantity as the decision variables of a mixture of backorders and lost sales inventory model in which the shortages are allowed and consider the demand of the lead time with normal distribution. Inventory is continuously reviewed, an order of size Q is made whenever the inventory level (based on the number of non-defective items) falls to the reorder point r . Besides, they assume that an arrival order may contain some defective units, and the number of defective units in an arrival order of size Q is a random variable with binomial probability distribution. In addition, we also assume that the purchaser inspects the entire items and the defective units in each lot will be returned to the vendor at

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the time of delivery of the next lot. Therefore, the model will have an extra cost in the inspection of each lot and an extra cost for holding the defective units in stock until the time they are returned to the supplier.

In this study, we seek to extend the model of Ouyang and Wu (1999) with the fixed rate propose a more general model that allows the backorder rate as a control variable. According to the opinion of Ouyang and Chuang (2001) (also see Lee (2005)), under most market behaviors, they think that many products of famous brands or fashionable commodities may lead to a situation in which customers prefer their demands to be backordered while shortages occur. Certainly, if the quantity of shortages is accumulated to a degree that exceeds the waiting patience of customers, some may refuse the backorder case. This phenomenon reveals that, as shortages occur, the longer the length of lead time is, the larger the amount of shortages is, the smaller the proportion of customers can wait, and hence the smaller the backorder rate would be. Hence, they assumed that the backorder rate is dependent on the length of lead time through the amount of shortages. So, this paper consider a continuous review inventory model with the backorder rate that proposed by using the idea of Ouyang and Chuang (2001) (also see Lee (2005)) to present a new general form.

However, when the demands of the different customers are not identical in the lead time, then we cannot use only a single distribution (such as Ouyang and Wu, 1999; Ouyang et al., 1999) to describe the demand of the lead time. Hence, we consider the mixtures of distribution (see Everitt and Hand, 1981). We first assume that the lead time demand follows a mixture of normal distribution and develop an algorithm procedure to find the optimal order quantity, the optimal lead time and the optimal reorder point. And then we consider any mixture of distribution function (d.f.), say $F_* = pF_1 + (1-p)F_2$, of the lead time demand has only known finite first and second moments (and hence, mean and variance are also known and finite) but make no assumption on the distribution form of F_* . That is, F_1 and F_2 of F_* belong to the class Ω of all single d.f.'s with finite mean and variance. Our goal is to solve a mixture inventory model with defective units by using the minimax distribution free approach. The minimax distribution free approach for our inventory model is to find the most unfavorable d.f.s F_1 and F_2 of F_* for each decision variable and then to minimize over the decision variables. Furthermore, the purpose of this paper is to develop an algorithm procedure for the mixture inventory model with defective units to find the optimal order quantity, optimal lead time and the optimal reorder point when the dis-

tribution of the lead time demand is mixtures of normal distribution or mixtures of free distribution. Finally, two numerical examples are also given to illustrate that when $p = 0$ or 1 , the model considers only one kind of customers' demand; when $0 < p < 1$, the model considers two kinds of customers' demand for the fixed fraction of the demand backordered β and defective rate θ . It implies that the minimum expected total annual cost of two kinds of customers' demand are larger than the minimum expected total annual cost of one kind of customers' demand. Thus, the minimum expected total annual cost increases as the distance between p and 0 (or 1) increases for the fixed β and θ . Hence, if the true distribution of the lead time demand is mixtures of normal distribution or mixtures of free distribution, we use a single distribution (such as Ouyang and Wu, 1999; Ouyang et al., 1999) to substitute the true distribution of the lead time demand; then the minimum expected total annual cost will be underestimated.

2. Notations and assumptions

To establish the mathematical model, the notations and non-negative cost parameters are exactly the same as those in Ouyang and Wu (1999) except the following notations:

- X the lead time demand which has a mixture of d.f. $F_* = pF_1 + (1-p)F_2$ with finite mean μ_*L and standard deviation $\sigma_*\sqrt{L}$ (> 0), a random variable.
- x^+ maximum value of x and 0 , i.e., $x^+ = \max\{x, 0\}$.
- x^- maximum value of $-x$ and 0 , i.e., $x^- = \max\{-x, 0\}$.
- r reorder point.
- $I_{(0 < X < r)} = \begin{cases} 1, & 0 < x < r, \\ 0, & \text{o.w.} \end{cases}$

The assumptions of the model are exactly the same as those in Ouyang and Wu (1999) except the following assumptions: The reorder point $r =$ expected demand during lead time + SS, and SS = k (standard deviation of lead time demand), i.e., $r = \mu_*L + k\sigma_*\sqrt{L}$, where $\mu_* = p\mu_1 + (1-p)\mu_2$, $\sigma_* = \sqrt{1+p(1-p)\eta^2}\sigma$, $\mu_1 = \mu_* + (1-p)\eta\sigma/\sqrt{L}$, $\mu_2 = \mu_* - p\eta\sigma/\sqrt{L}$ (i.e., $\mu_1 - \mu_2 = \eta\sigma/\sqrt{L}$, $\eta \in R$), and k is the safety factor which satisfies $p(X > r) = 1 - p\Phi(r_1) - (1-p)\Phi(r_2) = q$, where Φ represents the cumulative distribution function of the standard normal random variable, q represents the allowable stock-out probability during L , $r_1 = (r - \mu_1L)/(\sigma\sqrt{L}) = k\sqrt{1+p(1-p)\eta^2} - (1-p)\eta$, and $r_2 = (r - \mu_2L)/(\sigma\sqrt{L}) = k\sqrt{1+p(1-p)\eta^2} + p\eta$. Moreover, arrival order may contain some defective units. In addition, we also assume that the

number of non-defective units in an arriving order of size Q be a binomial random variable with parameters Q and $1 - \theta$, where θ ($0 \leq \theta \leq 1$) represents the probability of defect. Upon arrival of order, the entire items are inspected and defective units in each lot will be returned to the vendor at the time of delivery of the next lot.

3. Model formulation

3.1 The mixtures of normal distribution model

First we consider the lead time demand X has a mixture of normal distribution $F_* = pF_1 + (1-p)F_2$, where F_1 has a normal distribution with finite mean μ_1 and standard deviation $\sigma\sqrt{L}$ and F_2 has a normal distribution with finite mean μ_2 and standard deviation $\sigma\sqrt{L}$, $\mu_1 - \mu_2 = \eta\sigma/\sqrt{L}$, $\eta \in R$. Therefore, the demand of the lead time X has mixtures of probability density function (p.d.f.) which is given by

$$f(x) = p \frac{1}{\sqrt{2\pi}\sigma\sqrt{L}} e^{-\frac{1}{2}[(x-\mu_1)/\sigma\sqrt{L}]^2} + (1-p) \frac{1}{\sqrt{2\pi}\sigma\sqrt{L}} e^{-\frac{1}{2}[(x-\mu_2)/\sigma\sqrt{L}]^2},$$

where $\mu_1 - \mu_2 = \eta\sigma/\sqrt{L}$, $\eta \in R$, $x \in R$, $0 \leq p \leq 1$, $\sigma > 0$ (see Everitt and Hand, 1981).

Moreover, if $(\mu_1 - \mu_2)^2 < 27\sigma^2/(8L)$ or $|\eta| < \sqrt{27/8}$ for any $0 \leq p \leq 1$, then the mixture of normal distribution is a unimodal distribution. When $(\mu_1 - \mu_2)^2 > 4\sigma^2/L$ or $|\eta| > 2$, at least we can find a p ($0 \leq p \leq 1$) makes the mixtures of normal distribution to be a bimodal distribution. As mentioned earlier, we have assumed that shortages are allowed and the reorder point $r = \mu_*L + k\sigma_*\sqrt{L}$, where k , μ_* and σ_* are defined as above. Then the expected demand shortage at the end of the cycle is given by

$$B(r) = E[X - r]^+ = \int_r^\infty (x - r) dF_*(x) = \sigma\sqrt{L}\Psi(r_1, r_2, p), \quad (1)$$

where $\Psi(r_1, r_2, p) = p[\phi(r_1) - r_1(1 - \Phi(r_1))] + (1-p)[\phi(r_2) - r_2(1 - \Phi(r_2))]$, ϕ and Φ be the standard normal p.d.f. and cumulative distribution function (c.d.f.), respectively. Thus, the expected number of backorders per cycle is $\beta B(r)$ and the expected lost sales per cycle is $(1 - \beta)B(r)$, where β ($0 \leq \beta \leq 1$) is the fraction of the demand during the stock-out period being backorder. Hence, the stock-out cost per cycle is $[\pi + \pi_0(1 - \beta)]B(r)$.

Then, the expected net inventory level just before the order arrives is

$$\begin{aligned} & E[(X - r)^- I_{(0 < X < r)}] - \beta B(r) \\ &= \sigma\sqrt{L} \left\{ p \left[r_1 \Phi\left(\frac{\mu_*\sqrt{L}}{\sigma} + (1-p)\eta\right) - \phi\left(\frac{\mu_*\sqrt{L}}{\sigma} + (1-p)\eta\right) \right] + (1-p) \left[r_2 \Phi\left(\frac{\mu_*\sqrt{L}}{\sigma} - p\eta\right) - \phi\left(\frac{\mu_*\sqrt{L}}{\sigma} - p\eta\right) \right] \right\} + (1 - \beta)B(r) \end{aligned}$$

and the expected net inventory level at the beginning of the cycle, given that there are y non-defective items in an arriving order of size Q , is

$$\begin{aligned} & y + E[(X - r)^- I_{(0 < X < r)}] - \beta B(r) \\ &= y + \sigma\sqrt{L} \left\{ p \left[r_1 \Phi\left(\frac{\mu_*\sqrt{L}}{\sigma} + (1-p)\eta\right) - \phi\left(\frac{\mu_*\sqrt{L}}{\sigma} + (1-p)\eta\right) \right] + (1-p) \left[r_2 \Phi\left(\frac{\mu_*\sqrt{L}}{\sigma} - p\eta\right) - \phi\left(\frac{\mu_*\sqrt{L}}{\sigma} - p\eta\right) \right] \right\} + (1 - \beta)B(r) \end{aligned}$$

Therefore, the expected holding cost per cycle is

$$\begin{aligned} & h \frac{y}{D} \left\{ \frac{y}{2} + \sigma\sqrt{L} \left\{ p \left[r_1 \Phi\left(\frac{\mu_*\sqrt{L}}{\sigma} + (1-p)\eta\right) - \phi\left(\frac{\mu_*\sqrt{L}}{\sigma} + (1-p)\eta\right) \right] + (1-p) \left[r_2 \Phi\left(\frac{\mu_*\sqrt{L}}{\sigma} - p\eta\right) - \phi\left(\frac{\mu_*\sqrt{L}}{\sigma} - p\eta\right) \right] \right\} + (1 - \beta)B(r) \right\} \end{aligned}$$

If we let $L_0 = \sum_{j=1}^n b_j$ and L_i be the length of lead time with components $1, 2, \dots, i$ crashed to their minimum duration, then L_i can be expressed as $L_i = \sum_{j=1}^n b_j - \sum_{j=1}^i (b_j - a_j)$, $i = 1, 2, \dots, n$; and the lead time crashing cost $R(L)$ per cycle for a given that there are y non-defective items in an arriving order size Q is

$C(y)$ = ordering cost + non-defective holding cost + stock-out cost + defective holding cost + inspecting cost + lead time crashing cost

$$\begin{aligned}
&= A + h \frac{y}{D} \left\{ \frac{y}{2} + \sigma \sqrt{L} \left[p \left[r_1 \Phi \left(\frac{\mu_* \sqrt{L}}{\sigma} + (1-p)\eta \right) \right. \right. \right. \\
&\quad \left. \left. - \phi \left(\frac{\mu_* \sqrt{L}}{\sigma} + (1-p)\eta \right) \right] + (1-p) \left[r_2 \Phi \left(\frac{\mu_* \sqrt{L}}{\sigma} \right. \right. \right. \\
&\quad \left. \left. - p\eta - \phi \left(\frac{\mu_* \sqrt{L}}{\sigma} - p\eta \right) \right] \right] \right\} + (1-\beta)B(r) \Big\} + [\pi \\
&\quad + \pi_0(1-\beta)]B(r) + h' \frac{y}{D} (Q-y) + vQ + R(L).
\end{aligned} \tag{2}$$

The expected length of the cycle time and cycle cost under the lot of size Q are, respectively,

$$E(T | Q) = \frac{E(Y | Q)}{D} \tag{3}$$

and

$$\begin{aligned}
E(C | Q) &= A + \frac{h}{2D} E(Y^2 | Q) + \frac{h\sigma\sqrt{L}}{D} E(Y | Q) \\
&\quad \left\{ p \left[r_1 \Phi \left(\frac{\mu_* \sqrt{L}}{\sigma} + (1-p)\eta \right) - \phi \left(\frac{\mu_* \sqrt{L}}{\sigma} + (1-p)\eta \right) \right] \right. \\
&\quad \left. + (1-p) \left[r_2 \Phi \left(\frac{\mu_* \sqrt{L}}{\sigma} - p\eta \right) - \phi \left(\frac{\mu_* \sqrt{L}}{\sigma} - p\eta \right) \right] \right\} \\
&\quad + \frac{h}{D} (1-\beta)B(r)E(Y | Q) + [\pi + \pi_0(1-\beta)]B(r) \\
&\quad + \frac{h'Q}{D} E(Y | Q) - \frac{h'}{D} E(Y^2 | Q) + vQ + R(L).
\end{aligned} \tag{4}$$

Hence, the expected total annual cost is

$$\begin{aligned}
EAC^N(Q, L) &= \frac{E(C | Q)}{E(T | Q)} = \frac{AD}{E(Y | Q)} + \frac{h}{2} \frac{E(Y^2 | Q)}{E(Y | Q)} \\
&\quad + h\sigma\sqrt{L} \left\{ p \left[r_1 \Phi \left(\frac{\mu_* \sqrt{L}}{\sigma} + (1-p)\eta \right) - \phi \left(\frac{\mu_* \sqrt{L}}{\sigma} \right. \right. \right. \\
&\quad \left. \left. + (1-p)\eta \right) \right] + (1-p) \left[r_2 \Phi \left(\frac{\mu_* \sqrt{L}}{\sigma} - p\eta \right) \right. \right. \\
&\quad \left. \left. - \phi \left(\frac{\mu_* \sqrt{L}}{\sigma} - p\eta \right) \right] \right\} + h(1-\beta)B(r) + [\pi \\
&\quad + \pi_0(1-\beta)] \frac{DB(r)}{E(Y | Q)} + h'Q - h' \frac{E(Y^2 | Q)}{E(Y | Q)} \\
&\quad + \frac{DvQ}{E(Y | Q)} + \frac{D}{E(Y | Q)} R(L).
\end{aligned} \tag{5}$$

For a given lot of size Q , we assume that the number of non-defective units is a random variable (Y), which has a binomial distribution with parameter Q and $1-\theta$. That is, Y has the binomial p.d.f. as

$$p(y) = C_y^Q (1-\theta)^y \theta^{Q-y}, \tag{6}$$

where $y=0,1,2,\dots,Q$, and $0 \leq \theta \leq 1$.

In this case,

$$E(Y | Q) = Q(1-\theta) \tag{7}$$

and

$$E(Y^2 | Q) = Q(1-\theta)[\theta + Q(1-\theta)]. \tag{8}$$

Substituting Eqs. (1), (7) and (8) into Eq. (5), we get

$$\begin{aligned}
EAC^N(Q, L) &= \frac{AD}{Q(1-\theta)} + \frac{h}{2} [\theta + Q(1-\theta)] \\
&\quad + h\sigma\sqrt{L} \left\{ p \left[r_1 \Phi \left(\frac{\mu_* \sqrt{L}}{\sigma} + (1-p)\eta \right) \right. \right. \\
&\quad \left. \left. - \phi \left(\frac{\mu_* \sqrt{L}}{\sigma} + (1-p)\eta \right) \right] + (1-p) \right. \\
&\quad \left. \left[r_2 \Phi \left(\frac{\mu_* \sqrt{L}}{\sigma} - p\eta \right) - \phi \left(\frac{\mu_* \sqrt{L}}{\sigma} - p\eta \right) \right] \right\} \\
&\quad + h(1-\beta)\sigma\sqrt{L}\Psi(r_1, r_2, p) + [\pi + \pi_0(1-\beta)] \\
&\quad \frac{D\sigma\sqrt{L}\Psi(r_1, r_2, p)}{Q(1-\theta)} + h'\theta(Q-1) + \frac{Dv}{1-\theta} \\
&\quad + \frac{D}{Q(1-\theta)} R(L).
\end{aligned} \tag{9}$$

The parameter $(0 \leq \beta \leq 1)$ is treated as a constant; however, in the real market as unsatisfied demands occur, the longer the length of lead time is, the larger the amount of shortages is, the smaller the proportion of customers can wait, and hence the smaller the backorder rate would be. Therefore, in model (9), we consider the backorder rate β , as a decision variable instead of the constant case to accommodate the practical inventory situation. During the stock-out period, the backorder rate β , is variable and is a function of L through $B(r) = E[X-r]^+$. The larger the expected shortage quantity is, the smaller the backorder rate would be. Thus we define

$$\beta = \frac{\delta}{1 + \varepsilon B(r)}, \quad 0 \leq \delta \leq 1, \quad 0 \leq \varepsilon < \infty. \tag{10}$$

As the value of ε increases, the expected total annual cost becomes close to the complete lost case (i.e. $\beta \rightarrow 0$) for $\delta = 1$. Conversely, decreasing the value of ε , the expected total annual cost

will approach the complete backordered case (i.e. $\beta \rightarrow 1$) for $\delta = 1$.

By using Eq. (10), the model (9) is the expected annual cost of our new model reduced to

$$\begin{aligned} \text{Min}_{Q>0, L>0} \text{EAC}^N(Q, L) &= \frac{AD}{Q(1-\theta)} + \frac{h}{2}[Q(1-\theta) + \theta] \\ &+ h\sigma\sqrt{L} \left\{ p \left[r_1 \Phi \left(\frac{\mu_*\sqrt{L}}{\sigma} + (1-p)\eta \right) - \phi \left(\frac{\mu_*\sqrt{L}}{\sigma} \right) \right. \right. \\ &\left. \left. + (1-p)\eta \right] + (1-p) \left[r_2 \Phi \left(\frac{\mu_*\sqrt{L}}{\sigma} - p\eta \right) - \phi \left(\frac{\mu_*\sqrt{L}}{\sigma} - p\eta \right) \right] \right\} + h \left(1 - \frac{\delta}{1+\varepsilon B(r)} \right) B(r) \\ &+ \frac{D}{Q(1-\theta)} \left[\pi + \pi_0 \left(1 - \frac{\delta}{1+\varepsilon B(r)} \right) \right] B(r) \\ &+ h'\theta(Q-1) + \frac{Dv}{1-\theta} + \frac{D}{Q(1-\theta)} R(L), \end{aligned} \quad (11)$$

In order to find the minimum total expected annual cost, taking the first partial derivatives of $\text{EAC}^N(Q, L)$ with respect to Q and L in each time interval $[L_i, L_{i+1}]$, we obtain

$$\begin{aligned} \frac{\partial \text{EAC}^N(Q, L)}{\partial Q} &= -\frac{AD}{Q^2(1-\theta)} + \frac{h}{2}(1-\theta) \\ &- \frac{D}{Q^2(1-\theta)} \left[\pi + \pi_0 \left(1 - \frac{\delta}{1+\varepsilon B(r)} \right) \right] B(r) \\ &+ h'\theta - \frac{D}{Q^2(1-\theta)} R(L). \end{aligned} \quad (12)$$

and

$$\begin{aligned} \frac{\partial \text{EAC}^N(Q, L)}{\partial L} &= \frac{h\sigma}{2\sqrt{L}} \left\{ p \left[r_1 \Phi \left(\frac{\mu_*\sqrt{L}}{\sigma} + (1-p)\eta \right) - \phi \left(\frac{\mu_*\sqrt{L}}{\sigma} + (1-p)\eta \right) \right] \right. \\ &\left. + (1-p) \left[r_2 \Phi \left(\frac{\mu_*\sqrt{L}}{\sigma} - p\eta \right) - \phi \left(\frac{\mu_*\sqrt{L}}{\sigma} - p\eta \right) \right] \right\} + \frac{h\mu_*}{2} \left\{ p \left(r_1 + \frac{\mu_*\sqrt{L}}{\sigma} \right) \right. \\ &\left. + (1-p)\eta \right\} \phi \left(\frac{\mu_*\sqrt{L}}{\sigma} + (1-p)\eta \right) + (1-p) \\ &\times \left(r_2 + \frac{\mu_*\sqrt{L}}{\sigma} - p\eta \right) \phi \left(\frac{\mu_*\sqrt{L}}{\sigma} - p\eta \right) \left\} + \frac{hB(r)}{2L} \end{aligned}$$

$$\begin{aligned} &\times \left(1 - \frac{\delta}{1+\varepsilon B(r)} \right) + \frac{h\delta\varepsilon}{2L} \left(\frac{B(r)}{1+\varepsilon B(r)} \right)^2 + \frac{DB(r)}{2Q(1-\theta)L} \\ &\times \left\{ \frac{\pi_0\delta\varepsilon B(r)}{[1+\varepsilon B(r)]^2} + \left[\pi + \pi_0 \left(1 - \frac{\delta}{1+\varepsilon B(r)} \right) \right] \right\} \\ &- \frac{D}{Q(1-\theta)} c_i, \end{aligned} \quad (13)$$

where $B(r) = \sigma\sqrt{L}\Psi(r_1, r_2, p)$.

It is clear that for any given r_1, r_2 and p , we have $\Psi(r_1, r_2, p) > 0$. Hence, for fixed $L \in [L_i, L_{i+1}]$ we obtain

$$\begin{aligned} \frac{\partial^2 \text{EAC}^N(Q, L)}{\partial Q^2} &= \frac{2D}{Q^3(1-\theta)} \left\{ A + [\pi + \pi_0 \right. \\ &\times \left. \left(1 - \frac{\delta}{1+\varepsilon B(r)} \right) \right] B(r) + R(L) \left. \right\} > 0 \end{aligned}$$

then $\text{EAC}^N(Q, L)$ is convex in Q . However, we can also show that for fixed Q , $\text{EAC}^N(Q, L)$ is concave function of $L \in [L_i, L_{i+1}]$, if

$$\min \left\{ \frac{\mu_*\sqrt{L}}{\sigma} + (1-p)\eta, \frac{\mu_*\sqrt{L}}{\sigma} - p\eta \right\} > \sqrt{2}.$$

Therefore, for fixed Q , the minimum expected total annual cost will occur at the end points of the interval $[L_i, L_{i+1}]$. Setting Eq. (12) equal to zero and solving for Q , we have

$$\begin{aligned} Q &= \left\{ \frac{2D}{(1-\theta)[h(1-\theta) + 2h'\theta]} \cdot \left\{ A + [\pi + \pi_0 \right. \right. \\ &\times \left. \left. \left(1 - \frac{\delta}{1+\varepsilon B(r)} \right) \right] B(r) + R(L) \right\} \right\}^{\frac{1}{2}} \end{aligned} \quad (14)$$

where $B(r)$ and $R(L)$ as the above definitions. Thus, we can establish the following iterative algorithm to find the optimal lead time and the optimal order quantity.

Algorithm 1

Step 1 : For given $A, D, h, h', v, \sigma, \pi, \pi_0, \eta, p, q, \varepsilon, \delta, \theta, a_i, b_i, c_i$, and $i = 1, 2, \dots, n$.

Step 2 : Use the a_i, b_i and $c_i, i = 1, 2, \dots, n$, compute L_i and compute $B(r)$ using Eq. (1).

Step 3 : Given η, p and q by using the computer software Intel Visual Fortran V9.0 (Inclusive of IMSL) (2005) and the subroutine ZREAL from IMSL to solve k of the equation $1 - p\Phi(r_1) - (1-p)\Phi(r_2) = q$ where $r_1 = k\sqrt{1+p(1-p)\eta^2} - (1-p)\eta$ and $r_2 = k\sqrt{1+p(1-p)\eta^2} + p\eta$. Further, we obtain r_1 and r_2 .

Step 4 : For each $L_i, i = 1, 2, \dots, n$, compute Q_i by

using Eq. (14).

Step 5 : For each pair (Q_i, L_i) , compute the corresponding expected total annual cost $EAC^N(Q_i, L_i)$, $i = 1, 2, \dots, n$.

Step 6 : Find $\min EAC^N(Q_i, L_i)$. If $EAC^N(Q^*, L^*) = \min EAC^N(Q_i, L_i)$, then (Q^*, L^*) is the optimal solution.

3.2 The mixtures of distribution free model

The information about the form of the mixtures of d.f. of lead time demand is often limited in practices. In this subsection, we relax the restriction about the form of the mixtures of d.f. of lead time demand, i.e., we assume here that the lead time demand X has a mixture of d.f. $F^* = pF_1 + (1-p)F_2$, where F_1 has finite mean μ_1 and standard deviation $\sigma\sqrt{L}$ and F_2 has a finite mean μ_2 and standard deviation $\sigma\sqrt{L}$, $\mu_1 - \mu_2 = \eta\sigma/\sqrt{L}$, $\eta \in \mathbb{R}$, but make no assumption on the mixtures of d.f.'s form of F^* . Now, we try to use a minimax distribution free procedure to solve this problem. If we let Ω denote the class of all single c.d.f. (include F_1 and F_2) with finite mean and standard deviation then the minimax distribution free approach for our problem is to find the most unfavorable c.d.f.s F_1 and F_2 in Ω for each decision variable and then to minimize over the decision variables; that is, our problem is to solve

$$\min_{Q>0, L>0, F_1, F_2 \in \Omega} \max EAC^F(Q, L) \quad (15)$$

where

$$\begin{aligned} EAC^F(Q, L) &= \frac{AD}{Q(1-\theta)} + \frac{h}{2}[Q(1-\theta) + \theta] \\ &\quad + h[k\sigma_*\sqrt{L} + (1-\beta)B(r)] \\ &\quad + \frac{D}{Q(1-\theta)}[\pi + \pi_0(1-\beta)]B(r) \\ &\quad + h'\theta(Q-1) + \frac{Dv}{1-\theta} \\ &\quad + \frac{D}{Q(1-\theta)}R(L), \\ \beta &= \left\{ \frac{\delta}{1+\varepsilon B(r)} \right\} \text{ and } B(r) = E(X-r)^+. \end{aligned}$$

In addition, we need the following Proposition 1 to solve the above problem of the model (15).

Proposition 1. For F_1 and $F_2 \in \Omega$,

$$E(X_i - r)^+ \leq \frac{1}{2} \left\{ \sqrt{\sigma^2 L + (r - \mu_i L)^2} - (r - \mu_i L) \right\} \quad (16)$$

$i = 1, 2$,

where r.v.s X_i has a single distribution function F_i , $i = 1, 2$. Moreover, the upper bound (16) is tight.

So, by using $F^* = pF_1 + (1-p)F_2$ and Eq. (16), we obtain

$$\begin{aligned} B(r) &\leq \frac{1}{2} \left(-k\sqrt{1+p(1-p)\eta^2}\sigma\sqrt{L} \right) + \frac{p}{2}\sigma\sqrt{L} \\ &\quad \times \left\{ \sqrt{1 + \left[k\sqrt{1+p(1-p)\eta^2} - (1-p)\eta \right]^2} \right\} \\ &\quad + \frac{1-p}{2}\sigma\sqrt{L} \left\{ \sqrt{1 + \left[k\sqrt{1+p(1-p)\eta^2} + p\eta \right]^2} \right\} \end{aligned} \quad (17)$$

From the definition of the backorder rate β and inequality (17), we have

$$\begin{aligned} \beta &= \left\{ \frac{\delta}{1+\varepsilon B(r)} \right\} \\ &\geq \delta \cdot \left\{ 1 + \varepsilon \left\{ \frac{1}{2} \left(-k\sqrt{1+p(1-p)\eta^2}\sigma\sqrt{L} \right) \right. \right. \\ &\quad \left. \left. + \frac{p}{2}\sigma\sqrt{L} \left[\sqrt{1 + \left[k\sqrt{1+p(1-p)\eta^2} - (1-p)\eta \right]^2} \right] \right. \right. \\ &\quad \left. \left. + \frac{1-p}{2}\sigma\sqrt{L} \left[\sqrt{1 + \left[k\sqrt{1+p(1-p)\eta^2} + p\eta \right]^2} \right] \right\} \right\}^{-1} \end{aligned} \quad (18)$$

where $0 \leq \delta \leq 1$, $0 \leq \varepsilon \leq \infty$, $0 \leq p \leq 1$, $\sigma > 0$.

By using the inequality (17) and (18), the model (15) can be reduced to

$$\min_{Q>0, L>0} EAC^U(Q, L) \quad (19)$$

where

$$\begin{aligned} EAC^U(Q, L) &= \frac{AD}{Q(1-\theta)} + h \left[\frac{Q(1-\theta) + \theta}{2} \right. \\ &\quad \left. + k\sqrt{1+p(1-p)\eta^2}\sigma\sqrt{L} \right] + \left[\frac{D}{Q(1-\theta)} \pi \right. \\ &\quad \left. + \left(h + \frac{D}{Q(1-\theta)} \pi_0 \right) \cdot \left(1 - \frac{\delta}{1+\Delta^U(L)} \right) \right] \cdot \left(\frac{\Delta^U(L)}{\varepsilon} \right) \\ &\quad + h'\theta(Q-1) + \frac{Dv}{1-\theta} + \frac{D}{Q(1-\theta)}R(L), \end{aligned}$$

$$\Delta^U(L) = \varepsilon \cdot \left\{ \frac{1}{2} \left[-k\sqrt{1+p(1-p)\eta^2}\sigma\sqrt{L} \right] + \frac{p}{2}\sigma\sqrt{L} \left[\sqrt{1+\left[k\sqrt{1+p(1-p)\eta^2} - (1-p)\eta\right]^2} \right] + \frac{1-p}{2}\sigma\sqrt{L} \left[\sqrt{1+\left[k\sqrt{1+p(1-p)\eta^2} + p\eta\right]^2} \right] \right\}$$

and $EAC^U(Q, L)$ and $\Delta^U(L)$ are the least upper bounds of $EAC^F(Q, L)$ and $\Delta(L)$, respectively.

To minimize Eq. (19), we take the partial derivatives of $EAC^U(Q, L)$ with respect to Q and L , we obtain

$$\begin{aligned} \frac{\partial EAC^U(Q, L)}{\partial Q} &= -\frac{AD}{Q^2(1-\theta)} + \frac{h}{2}(1-\theta) \\ &- \frac{D}{Q^2(1-\theta)} \left[\pi + \pi_0 \left(1 - \frac{\delta}{1 + \Delta^U(L)} \right) \right] \cdot \left(\frac{\Delta^U(L)}{\varepsilon} \right) \\ &+ h'\theta - \frac{D}{Q^2(1-\theta)} R(L) \end{aligned} \quad (20)$$

and

$$\begin{aligned} \frac{\partial EAC^U(Q, L)}{\partial L} &= \frac{hk}{2\sqrt{L}} \left[\sqrt{1+p(1-p)\eta^2}\sigma \right] \\ &+ \frac{\delta}{2\varepsilon} \left(h + \frac{D}{Q(1-\theta)}\pi_0 \right) \left(\frac{[\Delta^U(L)]^2}{L[1 + \Delta^U(L)]^2} \right) \\ &+ h \left(1 - \frac{\delta}{1 + \Delta^U(L)} \right) \left(\frac{\Delta^U(L)}{2\varepsilon L} \right) \\ &+ \frac{D}{Q(1-\theta)} \left[\pi + \pi_0 \left(1 - \frac{\delta}{1 + \Delta^U(L)} \right) \right] \left(\frac{\Delta^U(L)}{2\varepsilon L} \right) \\ &- \frac{D}{Q(1-\theta)} c_i, \end{aligned} \quad (21)$$

where $\Delta^U(L) = \varepsilon \sigma \sqrt{L} \Psi(r_1, r_2, p) = \varepsilon B(r)$.

In addition, it can be verified that for fixed $L \in [L_i, L_{i+1}]$, $EAC^U(Q, L)$ is convex in Q . For fixed Q , $EAC^U(Q, L)$ is concave in $L \in [L_i, L_{i+1}]$. Therefore, for fixed Q , the minimum expected total annual cost will occur at the end points of the interval $L \in [L_i, L_{i+1}]$. Hence, setting Eq.(20) equal to zero and solving for Q , we get

$$\begin{aligned} Q &= \left\{ \frac{2D}{(1-\theta)[h(1-\theta) + 2h'\theta]} \times \left\{ A + \left[\pi \right. \right. \right. \\ &\quad \left. \left. + \pi_0 \left(1 - \frac{\delta}{1 + \Delta^U(L)} \right) \right] \left(\frac{\Delta^U(L)}{\varepsilon} \right) + R(L) \right\} \right\}^{\frac{1}{2}} \end{aligned} \quad (22)$$

Theoretically, for fixed $A, D, h, h', v, \sigma, \pi, \pi_0, \eta, p, q, \varepsilon, \delta, \theta$, and each $L_i (i=1, 2, \dots, n)$, the optimal (Q_i, L_i) pair given L_i can be obtained by solving Eq. (22) iteratively until convergence. The convergence of the procedure can be shown. Furthermore, using Eq. (19), we can obtain the corresponding expected total annual cost $EAC^U(Q_i, L_i)$. Hence, the minimum expected total annual cost is $\min_{i=0,1,\dots,n} EAC^U(Q_i, L_i)$. However, in practice, since the p.d.f. f_X of X is unknown, even if the value of q is given, we can not get the exact value of k . Thus, in order to find the value of k , we need the following proposition.

Proposition 2. Let Y be a random variable which has a p.d.f. $f_Y(y)$ with finite mean μL and standard deviation $\sigma\sqrt{L} (> 0)$, the for any real number $d > \mu L$,

$$P(Y > d) \leq \frac{\sigma^2 L}{\sigma^2 L + (d - \mu L)^2}. \quad (23)$$

So, by using $F_* = pF_1 + (1-p)F_2$, the recorder point $r = \mu_* L + k\sigma_*\sqrt{L}$ and the Proposition 2, we get

$$\begin{aligned} P(X > r) &\leq \frac{p}{1 + \left[k\sqrt{1+p(1-p)\eta^2} - (1-p)\eta \right]^2} \\ &+ \frac{1-p}{1 + \left[k\sqrt{1+p(1-p)\eta^2} + p\eta \right]^2} \end{aligned} \quad (24)$$

Further, it is assumed that the allowable stock-out probability q during lead time is given, that is, $q = P(X > r)$, then from Eq. (24), we get $0 \leq k \leq \sqrt{1/q-1} + |\eta|$.

It is easy to verify that $EAC^U(Q, L)$ has a smooth curve for $0 \leq k \leq \sqrt{1/q-1} + |\eta|$. Hence, we can establish the following algorithm to obtain the suitable k and hence the optimal Q and L .

Algorithm 2

Step 1 : For given $A, D, h, h', v, \sigma, \pi, \pi_0, \eta, p, q, \varepsilon, \delta, \theta, a_i, b_i, c_i$, and $i=1, 2, \dots, n$.

Step 2 : For a given q , we divide the interval $[0, \sqrt{1/q-1} + |\eta|]$ into m equal subintervals, where m is large enough. And we let $k_0 = 0$, $k_m = \sqrt{1/q-1} + |\eta|$ and $k_j = k_{j-1} + (k_m - k_0)/m$, $j=1, 2, \dots, m-1$.

Step 3 : Use the a_i, b_i and c_i , to compute L_i , $i=1, 2, \dots, n$.

Step 4 : For each L_i , $i=1, 2, \dots, n$, compute Q_i, k_j by using Eq. (22) for given $k_j \in \{k_0, k_1, \dots, k_m\}$, $j=0, 2, \dots, m$.

Step 5 : For each L_i , $i=1, 2, \dots, n$, compute the corresponding expected total annual cost

$EAC^U(Q_{i, k_j}, L_i),$

Step 6 : Find $\min_{k_j \in \{k_0, k_1, \dots, k_m\}} EAC^U(Q_{i, k_j}, L_i)$ and let $EAC^U(Q_{i, k_{s(i)}}, L_i) = \min_{k_j \in \{k_0, k_1, \dots, k_m\}} EAC^U(Q_{i, k_j}, L_i)$, then find $\min_{i=1,2,\dots,n} EAC^U(Q_{i, k_{s(i)}}, L_i)$. If $EAC^U(Q^*, L^*) = \min_{i=1,2,\dots,n} EAC^U(Q_{i, k_{s(i)}}, L_i)$, then (Q^*, L^*) is the optimal solution.

4. Numerical examples

In order to illustrate the above solution procedure and the effects of ordering cost reduction, let us consider an inventory system with the data: $D = 600$ units/year, $A = \$200$ per order, $h = \$20$, $h' = \$12$, $v = \$1.6$, $\pi = \$50$, $\pi_0 = \$150$, $\mu_* = 11$ units/week, $\sigma = 7$ units/week, $\theta = 0.00, 0.15, 0.30, 0.45, \varepsilon = 0, 2, 100, \infty$ (backorder case), $q = 0.2$ and the lead time has three components with shown in Table 1.

Table 1 Lead time data

Lead time component	Normal duration	Minimum duration	Unit crashing cost
i	$b_i(\text{days})$	$a_i(\text{days})$	c_i (\$/days)
1	20	6	0.4
2	20	6	1.2
3	16	9	5.0

Example 1. We assume here that the lead time demand follows a mixtures of normal distribution and want to solve the case when $\eta = 0.7$, $p = 0, 0.4, 0.8, 1.0$, $\varepsilon = 0, 2, 100, \infty$, $\theta = 0.00, 0.15, 0.30, 0.45$ and $\delta = 1.0, 0.6$. A summary of these optimal results is presented in (Q^*, L^*) and $EAC^N(Q^*, L^*)$ of Tables 2 - 3. From Tables 2 - 3, when $\eta = 0.7$, for the fixed p , defective rate θ and $\delta = 0.6, 1.0$, we can get the order quantity Q^* and the minimum expected total annual cost $EAC^N(Q^*, L^*)$ decrease as ε decreases (i.e. the backorder rate β increases) (also see Fig. 2). Moreover, for the fixed p , ε and $\delta = 0.6, 1.0$, the order quantity Q^* and the minimum expected total annual cost $EAC^N(Q^*, L^*)$ increase as θ increases (also see Fig. 1); for $\delta = 0.6, 1.0$, the fixed ε and defective rate θ , when $p = 0$ or 1, the model considers only one kind of customers' demand; when $0 < p < 1$, the model considers two kinds of customers' demand. It implies that $EAC^N(Q^*, L^*)$ of two kinds of customers' demand are larger than $EAC^N(Q^*, L^*)$ of one kind of customers' demand. Thus, the minimum expected total annual cost $EAC^N(Q^*, L^*)$ increases as the distance between p and 0 (or 1) increases for the fixed ε and defective rate θ (also see Fig. 3). In addition, no matter what values of p and θ , the optimal lead time L^* is approached to a certain value (3 weeks) for $\delta = 0.6, 1.0$ and $\varepsilon = 2, 100, \infty$. However, no matter what values of p and θ , the optimal lead time L^* is approached to a certain value (4 weeks) for $\delta = 0.6$ and $\varepsilon = 0$ and L^* is approached to a certain value (4 weeks) for $\delta = 1.0$ and $\varepsilon = 0$. Finally, the expected total annual cost $EAC^N(Q^*, L^*)$ is obtained by substituting Q^* and L^* into Eq. (11) when the d.f. of the lead time demand is the mixtures of normal distribution. The expected value of additional information (EVAI) is the largest amount that one is willing to pay for the knowledge of F_1 and F_2 and is equal to $EAC^N(Q^*, L^*) - EAC^N(Q^*, L^*)$. From Tables 2 - 3, we observe that for the fixed p and θ , EVAI decreases as ε increases (i.e. the backorder rate β decreases) except $\varepsilon = 0$, when $\delta = 0.6, 1.0$. Moreover, we also observe that for the given $\varepsilon = 2, 100, \infty$ and fixed θ , EVAI increases and then decreases as p increases when $\delta = 0.6, 1.0$. And the cost penalty $EAC^N(Q^*, L^*)/EAC^N(Q^*, L^*)$ of using the distribution free operating policy instead of the optimal one is decreasing as ε increases (i.e. the backorder rate β decreases) except $\varepsilon = 0$ fixed p and θ .

100, ∞ . However, no matter what values of p and θ , the optimal lead time L^* is approached to a certain value (4 weeks) for $\delta = 0.6, 1.0$ and $\varepsilon = 0$.

----- Insert Tables 2 - 3 here -----

----- Insert Figs. 1 - 3 here -----

Example 2. The data is as in Example 1. We assume here that the probability distribution of the lead time demand is mixtures of free. By using the proposed Algorithm 2 and $m = 500$ procedure yields these optimal results shown in (Q^*, L^*) and $EAC^U(Q^*, L^*)$ of Tables 2 - 3. From Tables 2 - 3, when $\eta = 0.7$, for the fixed p , defective rate θ and $\delta = 0.6, 1.0$, we can get the order quantity Q^* and the minimum expected total annual cost $EAC^U(Q^*, L^*)$ decrease as ε decreases (i.e. the backorder rate β increases) (also see Fig. 2). Moreover, for the fixed p , ε and $\delta = 0.6, 1.0$, the order quantity Q^* and the minimum expected total annual cost $EAC^U(Q^*, L^*)$ increase as θ increases (also see Fig. 1); the minimum expected total annual cost $EAC^U(Q^*, L^*)$ increases and then decreases as p increases for $\delta = 0.6, 1.0$ and the fixed ε and $\theta = 0, 0.15$ (also see Fig. 3). In addition, no matter what values of p and θ , the optimal lead time L^* is approached to a certain value (3 weeks) for $\delta = 0.6, 1.0$ and $\varepsilon = 2, 100, \infty$. However, no matter what values of p and θ , the optimal lead time L^* is approached to a certain value (3 weeks) for $\delta = 0.6$ and $\varepsilon = 0$ and L^* is approached to a certain value (4 weeks) for $\delta = 1.0$ and $\varepsilon = 0$. Finally, the expected total annual cost $EAC^N(Q^*, L^*)$ is obtained by substituting Q^* and L^* into Eq. (11) when the d.f. of the lead time demand is the mixtures of normal distribution. The expected value of additional information (EVAI) is the largest amount that one is willing to pay for the knowledge of F_1 and F_2 and is equal to $EAC^N(Q^*, L^*) - EAC^N(Q^*, L^*)$. From Tables 2 - 3, we observe that for the fixed p and θ , EVAI decreases as ε increases (i.e. the backorder rate β decreases) except $\varepsilon = 0$, when $\delta = 0.6, 1.0$. Moreover, we also observe that for the given $\varepsilon = 2, 100, \infty$ and fixed θ , EVAI increases and then decreases as p increases when $\delta = 0.6, 1.0$. And the cost penalty $EAC^N(Q^*, L^*)/EAC^N(Q^*, L^*)$ of using the distribution free operating policy instead of the optimal one is decreasing as ε increases (i.e. the backorder rate β decreases) except $\varepsilon = 0$ fixed p and θ .

5. Concluding remarks

In this article, we propose a continuous review inventory model by considering the mixtures of distribution of the lead time demand with defec-

tive units and controllable backorder rate to extend that of Ouyang and Chuang (2001) (also see Lee (2005)). In addition, we also develop an algorithmic procedure to find the optimal lead time. Further, we get the significant results in the expected total annual cost.

In future research on this problem, it would be interesting to deal with the inventory model with a service level constraint or treat the reorder point as a decision variable.

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Table 2 Summary of the optimal solution procedure (L_i in weeks and $\eta = 0.7, \delta = 1.0$)

ε	θ	(Q^*, L^*)	$EAC^U(Q^*, L^*)$	$EAC^N(Q^*, L^*)$	(Q_*, L_*)	$EAC^N(Q_*, L_*)$	EVAI	Cost penalty
$p = 0.0$ (or 1)								
∞	0.00	(169, 3)	5011.812	4754.263	(178, 3)	4749.207	5.057	1.0011
	0.15	(180, 3)	5521.167	5282.753	(190, 3)	5277.187	5.566	1.0011
	0.30	(196, 3)	6200.876	5987.094	(207, 3)	5980.871	6.223	1.0010
	0.45	(218, 3)	7172.743	6992.599	(230, 3)	6985.479	7.119	1.0010
100	0.00	(169, 3)	5006.326	4749.065	(178, 3)	4743.986	5.080	1.0011
	0.15	(180, 3)	5515.148	5277.051	(190, 3)	5271.459	5.592	1.0011
	0.30	(196, 3)	6194.171	5980.743	(206, 3)	5974.493	6.250	1.0010
	0.45	(218, 3)	7165.101	6985.362	(230, 3)	6978.211	7.151	1.0010
2	0.00	(159, 3)	4817.129	4557.497	(169, 3)	4552.247	5.250	1.0012
	0.15	(170, 3)	5307.551	5066.889	(180, 3)	5061.109	5.779	1.0011
	0.30	(185, 3)	5962.887	5746.644	(196, 3)	5740.184	6.460	1.0011
	0.45	(206, 3)	6901.468	6718.575	(218, 3)	6711.184	7.391	1.0011
0	0.00	(142, 4)	4185.610	3884.810	(134, 4)	3880.775	4.035	1.0010
	0.15	(150, 4)	4639.367	4323.690	(144, 4)	4320.613	3.077	1.0007
	0.30	(161, 4)	5243.830	4912.849	(156, 4)	4910.833	2.016	1.0004
	0.45	(177, 4)	6107.398	5761.479	(173, 4)	5760.489	0.990	1.0002
$p = 0.4$								
∞	0.00	(167, 3)	5018.595	4828.015	(181, 3)	4817.125	10.890	1.0023
	0.15	(179, 3)	5524.946	5362.674	(193, 3)	5350.686	11.988	1.0022
	0.30	(194, 3)	6200.787	6074.957	(210, 3)	6061.555	13.401	1.0022
	0.45	(216, 3)	7167.371	7091.307	(233, 3)	7075.976	15.331	1.0022
100	0.00	(167, 3)	5013.064	4822.918	(181, 3)	4811.979	10.939	1.0023
	0.15	(178, 3)	5518.877	5357.083	(193, 3)	5345.042	12.042	1.0023
	0.30	(194, 3)	6194.026	6068.730	(210, 3)	6055.269	13.461	1.0022
	0.45	(216, 3)	7159.666	7084.212	(233, 3)	7068.812	15.400	1.0022
2	0.00	(158, 3)	4824.652	4631.414	(171, 3)	4620.086	11.328	1.0025
	0.15	(169, 3)	5312.134	5147.003	(183, 3)	5134.533	12.470	1.0024
	0.30	(183, 3)	5963.687	5834.737	(199, 3)	5820.797	13.940	1.0024
	0.45	(204, 3)	6897.101	6817.562	(221, 3)	6801.615	15.948	1.0023
0	0.00	(142, 4)	4203.567	3918.194	(135, 4)	3915.470	2.723	1.0007
	0.15	(150, 4)	4656.681	4359.345	(145, 4)	4357.452	1.894	1.0004
	0.30	(161, 4)	5260.390	4951.497	(157, 4)	4950.432	1.065	1.0002
	0.45	(177, 4)	6123.053	5804.198	(175, 4)	5803.856	0.342	1.0001
$p = 0.8$								
∞	0.00	(168, 3)	5015.096	4787.362	(179, 3)	4779.563	7.799	1.0016
	0.15	(179, 3)	5522.304	5318.413	(191, 3)	5309.827	8.585	1.0016
	0.30	(195, 3)	6199.250	6026.050	(208, 3)	6016.453	9.597	1.0016
	0.45	(217, 3)	7167.342	7036.057	(231, 3)	7025.078	10.979	1.0016
100	0.00	(167, 3)	5009.578	4782.207	(179, 3)	4774.373	7.834	1.0016
	0.15	(179, 3)	5516.250	5312.758	(191, 3)	5304.135	8.624	1.0016
	0.30	(194, 3)	6192.505	6019.752	(208, 3)	6010.112	9.640	1.0016
	0.45	(216, 3)	7159.654	7028.881	(231, 3)	7017.853	11.028	1.0016
2	0.00	(158, 3)	4820.929	4590.640	(170, 3)	4582.543	8.096	1.0018
	0.15	(169, 3)	5309.249	5102.601	(181, 3)	5093.689	8.912	1.0017
	0.30	(184, 3)	5961.880	5785.666	(197, 3)	5775.703	9.962	1.0017
	0.45	(204, 3)	6896.768	6762.114	(219, 3)	6750.717	11.397	1.0017
0	0.00	(142, 4)	4195.115	3901.879	(135, 4)	3898.406	3.473	1.0009
	0.15	(150, 4)	4648.576	4341.640	(144, 4)	4339.120	2.520	1.0006
	0.30	(161, 4)	5252.694	4932.020	(156, 4)	4930.467	1.553	1.0003
	0.45	(177, 4)	6115.846	5782.344	(174, 4)	5781.662	0.682	1.0001

Table 3 Summary of the optimal solution procedure (L_i in weeks and $\eta = 0.7, \delta = 0.6$)

ε	θ	(Q^*, L^*)	$EAC^U(Q^*, L^*)$	$EAC^N(Q^*, L^*)$	(Q_*, L_*)	$EAC^N(Q_*, L_*)$	EVAI	Cost penalty
$p = 0.0$ (or 1)								
∞	0.00	(169, 3)	5011.812	4754.264	(178, 3)	4749.207	5.057	1.0011
	0.15	(180, 3)	5521.167	5282.753	(190, 3)	5277.187	5.567	1.0011
	0.30	(196, 3)	6200.876	5987.094	(207, 3)	5980.871	6.223	1.0010
	0.45	(218, 3)	7172.743	6992.599	(230, 3)	6985.480	7.119	1.0010
100	0.00	(169, 3)	5008.521	4751.146	(178, 3)	4746.075	5.070	1.0011
	0.15	(180, 3)	5517.557	5279.333	(190, 3)	5273.751	5.582	1.0011
	0.30	(196, 3)	6196.854	5983.285	(206, 3)	5977.045	6.240	1.0010
	0.45	(218, 3)	7168.160	6988.258	(230, 3)	6981.120	7.138	1.0010
2	0.00	(163, 3)	4896.300	4637.449	(172, 3)	4632.283	5.166	1.0011
	0.15	(174, 3)	5394.426	5154.605	(184, 3)	5148.919	5.686	1.0011
	0.30	(189, 3)	6059.679	5844.356	(200, 3)	5838.000	6.356	1.0011
	0.45	(211, 3)	7011.805	6829.937	(223, 3)	6822.666	7.271	1.0011
0	0.00	(156, 3)	4609.345	4295.486	(154, 4)	4283.872	11.614	1.0027
	0.15	(165, 3)	5090.929	4779.604	(164, 4)	4763.082	16.522	1.0035
	0.30	(177, 3)	5730.780	5427.273	(179, 4)	5403.986	23.287	1.0043
	0.45	(195, 3)	6641.947	6356.064	(199, 4)	6322.858	33.206	1.0053
$p = 0.4$								
∞	0.00	(167, 3)	5018.595	4828.015	(181, 3)	4817.125	10.890	1.0023
	0.15	(179, 3)	5524.946	5362.674	(193, 3)	5350.686	11.988	1.0022
	0.30	(194, 3)	6200.787	6074.957	(210, 3)	6061.556	13.401	1.0022
	0.45	(216, 3)	7167.372	7091.307	(233, 3)	7075.976	15.332	1.0022
100	0.00	(167, 3)	5015.278	4824.958	(181, 3)	4814.039	10.919	1.0023
	0.15	(179, 3)	5521.306	5359.321	(193, 3)	5347.300	12.021	1.0022
	0.30	(194, 3)	6196.732	6071.222	(210, 3)	6057.785	13.437	1.0022
	0.45	(216, 3)	7162.750	7087.051	(233, 3)	7071.679	15.372	1.0022
2	0.00	(162, 3)	4903.529	4711.271	(175, 3)	4700.134	11.137	1.0024
	0.15	(173, 3)	5398.690	5234.610	(187, 3)	5222.351	12.259	1.0023
	0.30	(188, 3)	6060.127	5932.321	(203, 3)	5918.617	13.704	1.0023
	0.45	(209, 3)	7007.040	6928.771	(226, 3)	6913.094	15.678	1.0023
0	0.00	(155, 3)	4621.130	4342.202	(156, 4)	4338.804	3.397	1.0008
	0.15	(165, 3)	5102.196	4830.267	(167, 4)	4822.121	8.146	1.0017
	0.30	(177, 3)	5741.465	5483.066	(181, 4)	5468.312	14.755	1.0027
	0.45	(194, 3)	6651.958	6418.864	(201, 4)	6394.403	24.460	1.0038
$p = 0.8$								
∞	0.00	(168, 3)	5015.096	4787.362	(179, 3)	4779.563	7.799	1.0016
	0.15	(179, 3)	5522.304	5318.413	(191, 3)	5309.828	8.585	1.0016
	0.30	(195, 3)	6199.250	6026.050	(208, 3)	6016.453	9.597	1.0016
	0.45	(217, 3)	7167.342	7036.057	(231, 3)	7025.078	10.979	1.0016
100	0.00	(168, 3)	5011.786	4784.270	(179, 3)	4776.450	7.820	1.0016
	0.15	(179, 3)	5518.673	5315.021	(191, 3)	5306.413	8.608	1.0016
	0.30	(195, 3)	6195.204	6022.273	(208, 3)	6012.650	9.623	1.0016
	0.45	(216, 3)	7162.731	7031.752	(231, 3)	7020.744	11.008	1.0016
2	0.00	(162, 3)	4899.896	4670.562	(174, 3)	4662.596	7.966	1.0017
	0.15	(173, 3)	5395.902	5190.283	(185, 3)	5181.514	8.769	1.0017
	0.30	(188, 3)	6058.427	5883.336	(201, 3)	5873.534	9.802	1.0017
	0.45	(209, 3)	7006.827	6873.427	(224, 3)	6862.213	11.214	1.0016
0	0.00	(156, 3)	4615.837	4317.087	(155, 4)	4309.793	7.294	1.0017
	0.15	(165, 3)	5097.175	4802.824	(165, 4)	4790.684	12.140	1.0025
	0.30	(177, 3)	5736.749	5452.563	(180, 4)	5433.750	18.814	1.0035
	0.45	(195, 3)	6647.590	6384.256	(200, 4)	6355.575	28.681	1.0045

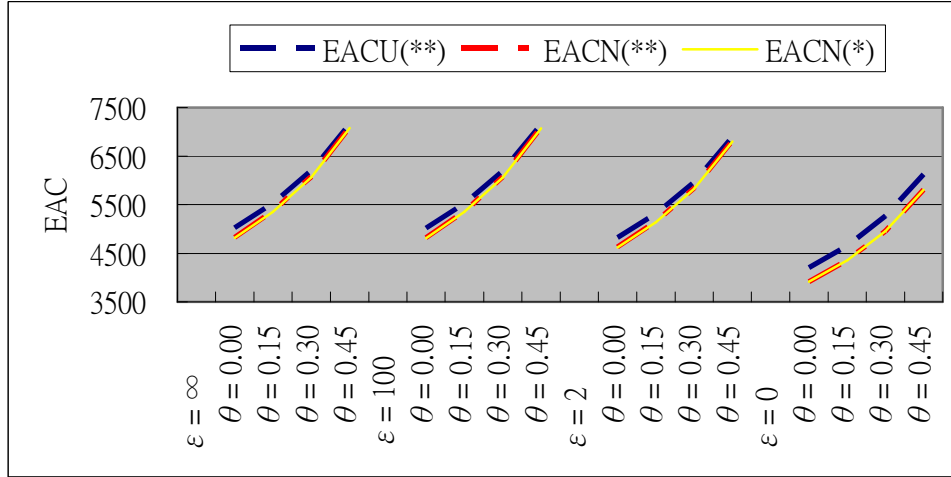


Fig. 1. Summary of the results of the optimal procedure based on $\eta = 0.7$, $p = 0.4$ and $\delta = 1.0$. Note: $EAC^U(Q^*, L^*)$, $EAC^N(Q^*, L^*)$ and $EAC^N(Q_*, L_*)$ will be denoted by the symbols EACU(**), EACN(**) and EACN(*), respectively.

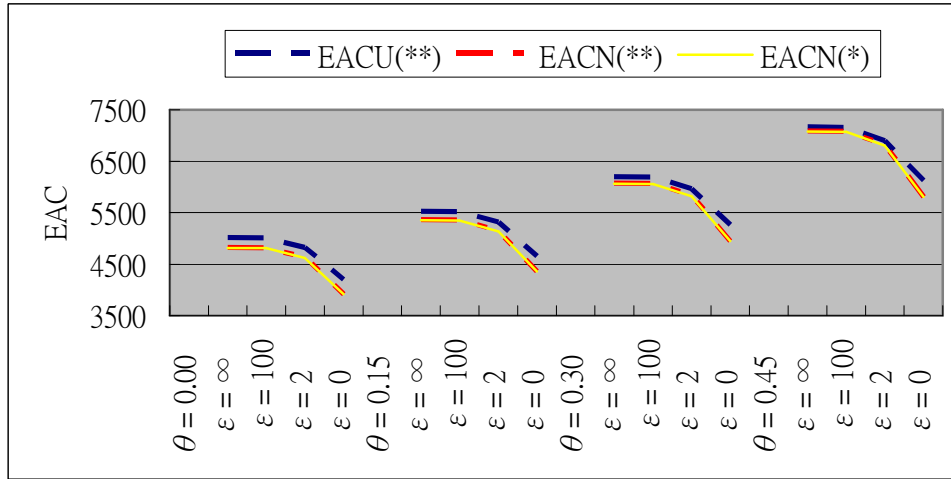


Fig. 2. Summary of the results of the optimal procedure based on $\eta = 0.7$, $p = 0.4$ and $\delta = 1.0$.

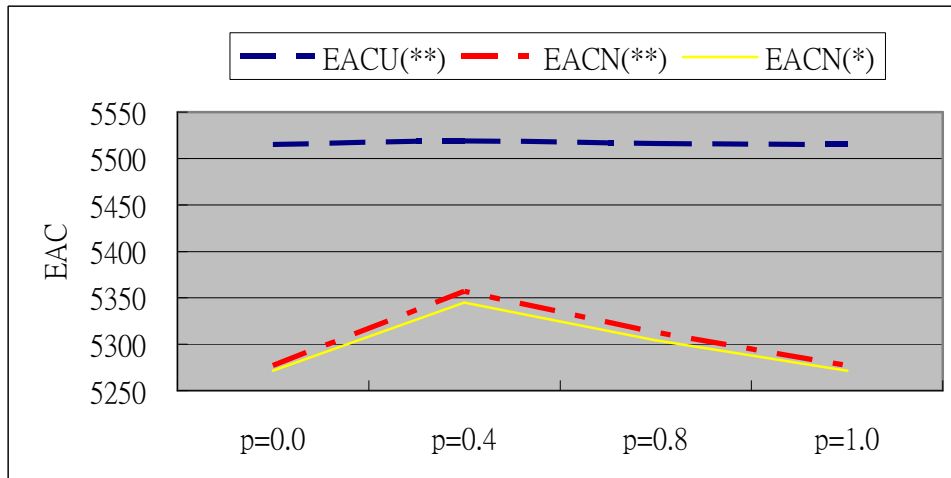


Fig. 3. Summary of the results of the optimal procedure based on $\eta = 0.7$, $\varepsilon = 100$, $\theta = 0.15$ and $\delta = 1.0$. Note: $EAC^U(Q^*, L^*)$, $EAC^N(Q^*, L^*)$ and $EAC^N(Q_*, L_*)$ will be denoted by the symbols EACU(**), EACN(**) and EACN(*), respectively.